

Decision-Sufficient Semantic Maps: Planner-Aware Compression for Goal-Directed Navigation

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Abstract—Open-vocabulary maps can store a semantic vector at every cell, point, or graph node, but a navigation policy rarely needs all of those vectors for its current task distribution. We study *decision-sufficient semantic mapping*: retaining the smallest task-conditioned subset of semantic landmarks that preserves downstream planner decisions. For a goal-selection planner that combines language similarity with geodesic cost, we score each landmark by the weighted frequency with which it is the full-map optimum on calibration state–query pairs, then retain the top landmarks under a memory budget. This selection is exactly optimal for weighted calibration-decision preservation under a cardinality budget. In controlled experiments over 30 procedural semantic topological environments, retaining only 25% of semantic landmarks preserves 72.6% of the full planner’s first actions, versus 61.0% for an embedding-confidence baseline, while reducing utility regret by 29.8%. The advantage persists across 10–40% budgets but disappears under a deliberately severe query-frequency shift, exposing a key limitation of task-conditioned compression. These results motivate semantic maps that optimize for downstream decisions rather than reconstruction fidelity alone.

I. INTRODUCTION

Semantic mapping has progressed from dense class-labeled reconstructions [1], [2] to scene graphs [3], [5] and language-indexable maps that support open-vocabulary navigation [4], [6]. Richer semantics make maps more useful, but storing high-dimensional language features densely can also become a memory bottleneck; recent work explicitly targets more compact language-map representations [7]. Existing map-compression work has likewise shown that what should be preserved can depend on the downstream path-planning objective [8].

We ask a narrower question: *if geometry is already available for collision-free planning, which semantic landmarks are actually necessary to reproduce the planner’s decisions?* This distinction matters because conventional confidence- or coverage-based thinning can retain semantically “good” landmarks that never alter a route, while discarding lower-confidence landmarks that are decisive for frequent goals.

We propose **Decision-Sufficient Semantic Mapping (DSSM)**, a planner-aware compression rule for topological semantic maps. DSSM observes a small calibration workload, assigns each landmark the weighted mass of decisions for which it is the full-map optimum, and keeps the top- B landmarks. Our contributions are: (1) a decision-preservation objective that directly links semantic-map memory to navigation behavior; (2) an exact top- B solution for the calibration objective, with a

simple preservation guarantee; and (3) a controlled evaluation that measures target agreement, first-action agreement, and utility regret across memory budgets and under distribution shift. We intentionally treat this as a representation study: the geometry is not compressed, and no claim is made that procedural graphs substitute for full embodied evaluation.

II. DECISION-SUFFICIENT SEMANTIC MAPPING

A. Planner and compression objective

Let a topological map contain landmarks $i \in \{1, \dots, N\}$ with positions x_i and unit semantic embeddings $z_i \in \mathbb{R}^d$. Geometry defines geodesic distance $d_G(s, x_i)$ from robot state s . For a unit language-query embedding q , we use a simple semantic goal selector

$$U_i(s, q) = q^\top z_i - \lambda \frac{d_G(s, x_i)}{D_G}, \quad (1)$$

$$i^*(s, q) = \arg \max_i U_i(s, q).$$

where D_G is graph diameter. This abstraction captures a common tension in semantic navigation: a landmark should be linguistically relevant without being arbitrarily costly to reach. Once a target is selected, the robot follows a shortest path to it.

Given a calibration set $\mathcal{C}$ of state–query pairs, define the full-map margin

$$m(s, q) = U_{i^*}(s, q) - \max_{j \neq i^*} U_j(s, q) \quad (2)$$

and landmark decision impact

$$I_i = \sum_{(s, q) \in \mathcal{C}} \mathbf{1}[i = i^*(s, q)] (1 + \gamma m(s, q)). \quad (3)$$

We use $\gamma = 0.5$. DSSM retains the B landmarks with largest I_i . To reduce brittleness to the observed workload, 20% of calibration queries are uniform *coverage probes*; the remainder follow the observed query distribution.

B. Why top- B is the right calibration solution

Assume a unique full-map maximizer for each calibration pair. For any retained subset S , the compressed planner maximizes the same utility (1) over $i \in S$. A calibration decision is therefore preserved if and only if its full-map winner $i^*(s, q)$ is retained: if it is retained, no other retained

node can outrank the full-map maximizer; if it is removed, that target cannot be selected.

Consequently, the weighted number of preserved calibration decisions is

$$J(S) = \sum_{i \in S} I_i. \quad (4)$$

Under $|S| \leq B$, selecting the B largest impacts exactly maximizes $J(S)$. In particular, the union of all calibration winners is a decision-sufficient set that preserves every calibration target. DSSM is the budgeted version when that set is too large. This guarantee concerns the calibration workload; generalization depends on how representative its states and queries are.

III. CONTROLLED EVALUATION

A. Protocol

We evaluate 30 independently generated connected topological environments. Each contains $N = 120$ landmarks positioned in $[0, 1]^2$ and connected by an undirected 6-nearest-neighbor graph. Each landmark has a 24-D semantic vector generated around one or two of 12 latent concept prototypes, with heterogeneous observation confidence and noise. Query concepts follow a Zipf-like workload distribution $p(k) \propto k^{-0.75}$, with additive embedding noise and 15% compound queries. We set $\lambda = 0.25$ in (1).

Per environment, DSSM is calibrated on 36 robot states and 90 queries, of which 20% are uniform coverage probes. Evaluation uses disjoint sets of 72 states and 160 queries. We sweep semantic-memory retention from 10% to 40%. All methods retain the complete geometric graph; only landmark semantic vectors are thinned.

We compare against: **Confidence**, retaining nodes with largest observation-confidence times prototype similarity; **Spatial k -center**, prioritizing geometric coverage; **Semantic k -center**, prioritizing embedding-space coverage; and **Random**. We report (i) *target agreement* with the full semantic map, (ii) *first-action agreement*, i.e., whether the next shortest-path hop matches the full-map planner, and (iii) utility regret $U_{i^*} - U_i$. Reported confidence intervals are 95% normal intervals over the 30 independent environments. The experiment script and master seed 20260810 are included with the source package.

B. Results

Figure 1 shows that planner-aware selection improves behavioral preservation across all budgets. At 25% retention, DSSM achieves $72.6\% \pm 1.3\%$ first-action agreement, compared with $61.0\% \pm 1.5\%$ for Confidence. Because environments are paired across methods, the DSSM–Confidence first-action gain is 11.6 ± 1.7 percentage points (95% CI). Target agreement rises from 46.0% to 62.3%, while mean utility regret falls from 0.0600 to 0.0421, a relative reduction of 29.8%.

The distinction between target and first-action agreement is useful: two semantic targets can disagree while inducing the same immediate route choice. Thus, map compression can preserve more of the robot’s local behavior than target identity alone suggests.

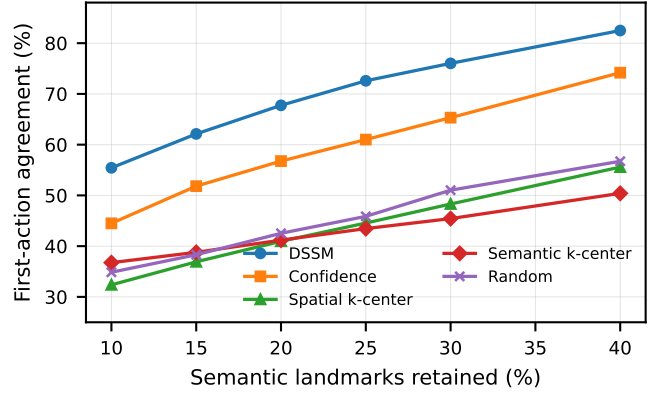


Fig. 1. First-action agreement with the full semantic-map planner. DSSM is strongest at every tested in-distribution memory budget; values are means over 30 independently generated environments.

TABLE I
RESULTS AT 25% SEMANTIC-MEMORY RETENTION. ID IS THE HELD-OUT IN-DISTRIBUTION WORKLOAD; SHIFT REVERSES THE CONCEPT-FREQUENCY ORDERING. MEANS OVER 30 ENVIRONMENTS.

Method	ID action $\uparrow$	ID regret $\downarrow$	Shift action $\uparrow$
DSSM	0.726	0.042	0.616
Confidence	0.610	0.060	0.622
Spatial k -center	0.446	0.116	0.446
Semantic k -center	0.435	0.105	0.428
Random	0.459	0.117	0.453

C. Distribution shift and limitations

Task conditioning creates an expected failure mode. We deliberately reverse the concept-frequency ordering at test time while keeping the map fixed. At 25% retention, DSSM drops to 61.6% first-action agreement and is statistically indistinguishable from the confidence baseline at 62.2% (Table I). Coverage probes prevent the retained set from depending exclusively on frequent queries, but they do not eliminate severe shift. A practical system should therefore update impact scores online, reserve a task-agnostic memory floor, or trigger re-expansion when the query distribution changes.

Our study has three additional limitations. First, semantics are procedural rather than produced by a deployed vision-language encoder. Second, DSSM compresses semantic memory but leaves geometry untouched. Third, Eq. (1) is deliberately simple; learned or hierarchical planners may have more complex decision dependencies. These constraints make the experiment diagnostic rather than a claim of end-to-end state of the art.

IV. CONCLUSION

DSSM reframes semantic-map compression as preservation of downstream navigation decisions. Under a fixed cardinality budget, decision-impact ranking exactly optimizes weighted calibration-decision preservation, and controlled experiments show a substantial in-distribution gain over confidence, semantic-diversity, and random thinning. The severe-shift

result is equally informative: a semantic map can be compact because it is adapted to a workload, but that compactness must be managed as the workload changes. Extending decision-sufficient mapping to online updates, learned planners, and embodied open-vocabulary benchmarks is the next step.

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